

MTH422A Assignment 1

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2024-01-31

Question 3

If the given function $f(x)$ is a valid pdf, it should satisfy these conditions:

1. $f(x) \geq 0$ for all x belonging to support of X .
2. Integration of $f(x)$ over support of X is equal to 1.

Bessel function is non-negative for all x in $(0, \text{Inf})$. Now, we integrate the function over support of X $(0, \text{Inf})$.

```
func<-function(x)
{
  return(2*besselK(2*sqrt(x), nu = 0))
}

integrate(func,0,Inf)
```

```
## 1 with absolute error < 8.1e-06
```

The answer is 1.

So, we can say that this function is a valid pdf.